

The following are excerpts from the Department-wide newsletter that are written or about students from the department. For the full newsletter, contact the department.

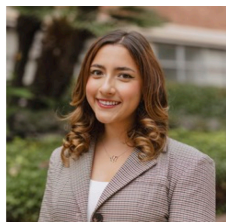
STUDENT EXPERIENCE

Biostatistics Summer Internships

Every summer, students in our department pursue summer internship opportunities across biotech, defense, technology, pharmaceuticals, and more! A panel of UCLA Biostatistics students share their experiences regarding internship search, application process, and reflections on experiences.

By **Cindy J. Pang**

The Panelists



Karen Harrison
MDSH
Centrillion Technologies



Julian Gavino
MS
Lockheed Martin



Siddhant Patel
MS
ASML



Darren Lin
PhD
AbbVie



Yanlong Bai
PhD
Genentech

How did you find your internship?

LinkedIn, Company Websites, getting referrals, leveraging alumni connections, and Handshake.

Walk us through your application timeline.

Patel: “I applied sometime between November through January ... [In] Total I spent 3-5 hours a week at most between browsing LinkedIn and applying.”

Bai: “I started my application from September or October and got the first interview in early February next year... I still recommend starting it [application] as early as possible. I spent about 2-3 hours/day to search for openings and have filled out over 100 application forms.”

Gavino: “I started in late September and got my offer in mid-April, but I would do a little bit of searching every day.”

Lin: “I applied sometime between November through January... I applied to a few internships over the course of a couple weeks.”

Harrison: “I started my summer internship search the beginning of January and was sending applications until May. My strategy was to stress the quality ... over quantity. I created a detailed recruitment spreadsheet. ... Over the course of 5 months, I applied to 103 positions, received 3 independent interviews, and one offer.”

What did you do to prepare for your interviews? Is there anything you wish you did differently?

Patel: “I leveraged my connections and my network’s connections, scheduled coffee chats with as many people as I could in the field most relevant to what I wanted to pursue. ... [If I did anything differently,] I would say start earlier.”

Bai: “1) Understand what titles or roles are relevant, and what knowledge or skillset they are looking for. Different tracks may have different requirements, e.g., data scientists/engineer track [is different from] biostatistics track. So make good use of AI tools for information gathering. 2) Optimize and refine your resume. If you want to apply for multiple roles, prepare role-specific resumes. 3) Practice for your interviews ... and be confident.”

Gavino: “I become a stronger storyteller during my interviews when talking about my experiences and wish I had a few more stories prepared beforehand.”

Lin: “For the interviews themselves, my LinkedIn and website helped visibility and giving the managers a sense of who I was before I met with them... I was ready to talk about my research interests and my goals. I was very honest about my intent with doing an internship, which was that I was still trying to figure out whether or not I wanted to do into academia or industry after graduation. ... I wish I had applied to more tech-heavy jobs (i.e., Google, Apple, etc.) and jobs in other fields.”

Harrison: “I watched data science YouTube videos to help prepare for AI and ML topics (StatQuest with Josh Starmer is a great creator). For each interview, I created Docs and took detailed notes on the company. I practiced recording myself and answering out-loud example interview questions that I would cater toward the specific position. I would then iterate by watching back the recording and retook it if I wanted my answer to sound more refined. ... One thing I’d do different is apply to more jobs so I could have more options ideally and interview practice.”

What surprised you most about the internship itself – whether about the work, culture, day-to-day rhythm, etc?

Patel: “To be honest, I thought my schedule would look like a typical 8-5... [but] I do my work, deliver my results, and network, but I’m also spending time playing pool with other interns and exploring the office campus.”

Bai: “The working pace in industry is pretty fast and compact. My calendar was generally filled up with meetings and my manager and collaborator usually expected to have some progress every week or every other week. ... Regarding career prospects, before this internship, I had heard that advanced degrees could play an important role in long-term career development in the pharmaceutical industry, but I did not have a clear sense of what that meant in practice. Through this experience, I gained a better understanding of the different career paths within the industry and saw that, particularly for more research-oriented or scientific leadership roles, having a PhD can be a valuable advantage and is quite common among people who advance in these areas.”

Gavino: “I was surprised about the amount of freedom I was given to complete my projects.”

Lin: “I was surprised by how organized AbbVie is. Everything is centralized and incredibly streamlined... through Microsoft Teams, it was incredibly easy to get in touch with other employees, set up meetings, book rooms, etc... The biggest surprise was the difference in what industry emphasized as important. Academia is very focused on cutting-edge research, with a strong emphasis on AI and more niche methods of statistics. In industry however, I learned that they place a very strong emphasis on popular interpretable models instead. ... Good presentations are king (i.e., Beamer is frowned upon and dense text-heavy

slides are discouraged)”

Harrison: “One of the most surprising parts of my internship experience is how independent you become, and confident you must be... I am there to learn and grow my own skillset, which can feel scary at times, but it is the best experience for growth.”

What is one piece of advice you’d give to a student who is just starting to think about internships?

Patel: “It’s a numbers game. Your resume needs to stand out, and what that looks like for everyone is different, but apply to anything and everything, if it’s moderately related to what you want long term. Pay doesn’t matter so much for an internship; it’s all about the experience and what you can speak to.”

Bai: “Think early, move early, and do NOT give up easily.”

Gavino: “Prioritize networking over applying online. For example, finding recruiters through campus events would be very helpful.”

Lin: “Do an internship early. It will help immensely with figuring out what you want to do after graduation. It also gives you a much wider perspective on how statistics is used broadly. For instance, I’ve learned what academia prioritizes is very different from what industry prioritizes. I’ve also learned a lot of skills and information that are hard to pick up on in the day-to-day of school (e.g. making good presentations, potential FDA policy changes, networking, how companies in pharma rank against each other, etc.). These things have actually given me some ideas on the direction to take my own research. It’s been great being able to step back a bit just to get some perspective on the field.”

Harrison: “Start. Early. I cannot emphasize this enough. If I were to go back and do it all over, I would start looking for internships throughout the entire school year, even the summer prior, just to send out applications early, and save dates when you know certain companies will be opening up their applications so you don’t miss the deadline. Even just dedicating 1 hour a week to recruitment starting in September will pay DIVIDENDS.”

If you have any questions for the students featured in this article, please feel free to email them or connect with them on LinkedIn!

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RESEARCH HIGHLIGHTS

Student Publication Highlights

Recent publications and preprints · Updated September 2026

Dissertation research

Kyle Lin Wu publishes two first-author papers

Kyle Lin Wu has published two first-author papers from his dissertation research with Professor Sudipto Banerjee. The papers address related challenges in spatial biostatistics: identifying meaningful health disparities between neighboring regions and understanding spatial confounding in multivariate disease analyses.

Assessing Spatial Disparities: A Bayesian Linear Regression Approach

Kyle Lin Wu and Sudipto Banerjee. *Biostatistics*.

DOI: [10.1093/biostatistics/kxaf048](https://doi.org/10.1093/biostatistics/kxaf048) | [arXiv: 2407.19171](https://arxiv.org/abs/2407.19171)

This paper develops a Bayesian linear regression framework for detecting sharp differences between neighboring geographic regions on disease maps. It uses posterior probabilities and Bayesian false discovery rate control to identify difference boundaries while retaining analytical tractability. The authors evaluate the method on maps of U.S. counties and apply it to county-level estimates of lung cancer mortality.

Spatial Confounding in Multivariate Areal Data Analysis

Kyle Lin Wu and Sudipto Banerjee. *Biometrics*, Volume 82, Issue 3, September 2026; published online August 11, 2026.

DOI: [10.1093/biomtc/ujag143](https://doi.org/10.1093/biomtc/ujag143) | [arXiv: 2505.07232](https://arxiv.org/abs/2505.07232)

This paper studies spatial confounding when several related outcomes are analyzed jointly. Using a Bayesian coregionalized areal regression model, the authors ex-

amine the issue from both analysis-model and data-generation perspectives. Their results show that multivariate spatial models can limit posterior variance inflation, need not add bias, and can improve precision over non-spatial alternatives, even when the spatial structure is misspecified.

Preprint

Filippo Monti leads new preprint

UCLA Biostatistics doctoral student Filippo Monti leads a new preprint with Andrew Holbrook, Nathan E. Glatt-Holtz, and Marc A. Suchard on efficient inference for multivariate Ornstein–Uhlenbeck processes.

Stable Matrix Parametrizations and Structured Adjoints for Ornstein–Uhlenbeck Processes

Filippo Monti, Andrew Holbrook, Nathan E. Glatt-Holtz, and Marc A. Suchard. *arXiv preprint*; submitted August 17, 2026.

DOI: [10.48550/arXiv.2608.16401](https://doi.org/10.48550/arXiv.2608.16401) | [arXiv: 2608.16401](https://arxiv.org/abs/2608.16401)

The preprint develops a stable and computationally efficient parametrization for multivariate Ornstein–Uhlenbeck processes. Estimation is challenging because the drift matrix must remain Hurwitz stable, while likelihood and gradient calculations repeatedly require matrix exponentials and Lyapunov-equation solutions. The proposed parametrization represents the drift through stable one- and two-dimensional blocks, enforces stability through scalar constraints, and supports smooth transitions between real- and complex-eigenvalue regimes. Structured adjoints accelerate likelihood and derivative calculations, with applications to asynchronous financial data and multivariate phylogenetic traits.

Letter from **Biostatistics Student Association (BSA)** Leadership



Cindy J. Pang



Nena Ojukwu

Dear Biostatistics Community,

We are pleased to welcome new and returning members of our Biostatistics community this fall quarter. To our new students: welcome to UCLA — and if you're new to the city, welcome to Los Angeles!

We also want to take a moment to celebrate the commitment and energy that our community showed this past year. Thanks to the continued support of the Department of Biostatistics and the Master of Data Science in Health (MDSH) program, our Lunch and Learn series and Alumni Panel were both great successes. These events brought together students, faculty, and alumni in ways that strengthened our shared sense of community. We're proud of the momentum these events built, and we're looking forward to continuing them, along with new programming throughout this year.

None of this happens without the dedication of our student volunteers, and we're grateful to everyone who has given their time to plan events and support one another beyond the classroom. Behind the scenes, our board works closely with department faculty and staff to keep these traditions running smoothly, and we thank them for their continued guidance.

BSA is always looking for new members to help shape our events and community initiatives. If you're interested in getting involved, please reach out to any current board member. We are also recruiting for several open board positions this year (details to follow soon, so keep an eye out via your Student Affairs Officers (SAO) and BSA social media channels)

We welcome all members of our community to connect with us anytime — in person, by email, or through our BSA social media channels. We look forward to a great year together!

Warm regards,
Cindy J. Pang & Nena Ojukwu
BSA Co-Presidents

BSA Board Members 2026-2027



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**Follow BSA Socials to be in the know
about events, townhalls, and more!**



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COMMUNITY HIGHLIGHTS

A Summer of Visibility and Recognition

From ISBA in Nagoya to JSM in Boston, UCLA Biostatistics faculty, students, alumni, and programs were active across the statistical sciences community this summer.

Compiled by the newsletter editor

Conference

UCLA Biostatistics at JSM

The Department of Biostatistics had a strong presence at the 2026 Joint Statistical Meetings in Boston, with 22 affiliated presentations by students, faculty, and collaborators throughout the week. The breadth of the program reflected the department's active role across methodological, applied, and interdisciplinary areas of statistical science.

UCLA Biostatistics and UCLA Statistics & Data Science hosted a very well-attended joint reception at the Westin Boston Seaport. The event brought together current students, faculty, alumni, and friends of the two departments for a warm evening of conversation and reconnection.

Invited Lecture

Marc Suchard recognized by WNAR

Professor Marc Suchard delivered the Western North American Region 2026 Research Impact Award Lecture at JSM, "Inference for Discrete-Valued Stochastic Processes in Infectious Diseases." The lecture highlighted the impact of his work at the intersection of statistical methodology, computation, and infectious disease research.

Student Award Session

Emma Landry recognized at JSM

After defending her dissertation in spring 2026, Dr. Emma Landry presented her work on warped functional mixed membership models for EEG spectral features in one of the SBSS Student Paper Award sessions. Her presentation showcased innovative statistical modeling for complex neurological data.

Program Visibility

MDSH reaches a global audience

The Master of Data Science in Health program hosted a table at the ISBA World Meeting, creating an opportunity to introduce the program to an international audience of researchers, students, and collaborators interested in Bayesian statistics, data science, and health.



Figure 6: Dr. Emma Landry holding the SBSS Travel Award Certificate and her PhD advisor, Dr. Donatello Telesca.

International Meeting

UCLA Biostatistics at ISBA

Earlier in the summer, our community was also well represented at the ISBA World Meeting in Nagoya, Japan, the biennial flagship conference of the International Society for Bayesian Analysis. Faculty and students presented their work and contributed to a week of scientific exchange with the international Bayesian community.

Grant

Andrew Holbrook receives NIGMS MIRA R35

Professor Andrew Holbrook received an NIGMS Maximizing Investigators' Research Award for Early Stage Investigators, providing \$2.1 million in total funding for his research program, "Deep Bayesian Biostatistics: AI-Accelerated Inference for Big N and Big P." The award provides long-term, flexible support for an ambitious research program at the interface of Bayesian statistics and

artificial intelligence.

Recognition

Soumyakanti Pan honored

Dr. Soumyakanti Pan, a recent graduate of our doctoral program and now a postdoctoral fellow at the University

of Washington, received the Dean's Outstanding Student Award in Biostatistics and was recently featured by the UCLA Fielding School of Public Health. The main article from his dissertation, published in Bayesian Analysis under the guidance of Professor Sudipto Banerjee, was also selected as one of three journal highlight papers at the ISBA World Meeting.

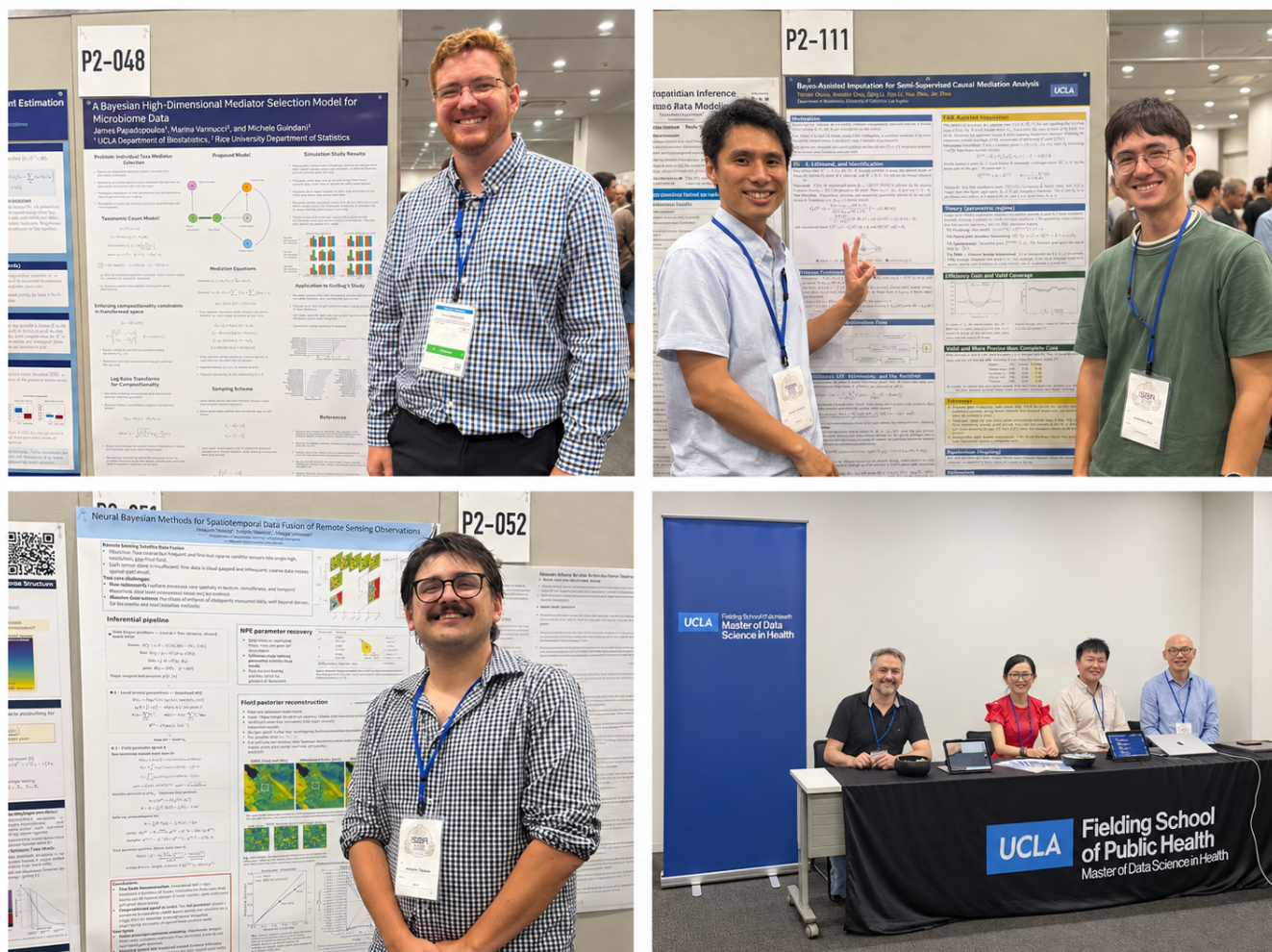


Figure 7: UCLA Biostatistics students presenting their research and MDSH faculty connecting with attendees at the 2026 ISBA World Meeting in Nagoya, Japan.

Seminars

UCLA Biostatistics seminars: <https://tiny.cc/UCLAbiostatseminars>

UCLA Statistics seminars: <https://tiny.cc/UCLASestatseminars>